

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is:

The logs show that: 203.0.113.0 sent messages to the server around 3 times a second preventing legitimate connections.

This event could be: DOS attack SYN Flood attack

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake:

- The [SYN] packet is the initial request from an employee visitor trying to connect to a web page hosted on the web server. SYN stands for "synchronize."
- The [SYN, ACK] packet is the web server's response to the visitor's request agreeing to the connection. The server will reserve system resources for the final step of the handshake. SYN, ACK stands for "synchronize acknowledge."
- The [ACK] packet is the visitor's machine acknowledging the permission to connect. This is the final step required to make a successful TCP connection. ACK stands for "acknowledge."

Explain what happens when a malicious actor sends a large number of SYN packets all at once: The server gets overloaded and unable to respond to requests.

Explain what the logs indicate and how that affects the server: The logs indicate that this is a Dos syn flood attack originating from a single source simulating TCP connections and flooding with syn packets.